Information, Experience, and Adaptation: Evidence on Climate Belief Persistence among Bangladeshi Adolescents *

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Abstract

Understanding how young people form and revise their perceptions of climate change is central to climate adaptation, particularly in climate-vulnerable countries where today's adolescents will bear a disproportionate share of future climate risks. This paper introduces a novel longitudinal dataset tracking more than 5,000 Bangladeshi students from 8th to 9th grade across three survey waves over a 14-month period. The dataset provides unusually rich measures of climate knowledge, beliefs, and attitudes (including concrete measures of willingness to pay to help address climate change), linked with student and teacher characteristics. We characterize climate-related attitudes and examine their persistence, leveraging two sources of exogenous variation: (i) a randomized climate-related education intervention and (ii) localized exposure to a major rainfall event. Across both empirical analyses, we find little evidence that light-touch information campaigns or recent climate shock exposure shifts climate knowledge, beliefs, or engagement. Climate-related outcomes are highly persistent across waves and are strongly associated with baseline quiz performance in math and climate science.

Key Words: climate beliefs, climate education, climate change, randomized controlled trial, rainfall shocks, belief updating

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1 Introduction

Building a climate-aware society is central to the challenge of adaptation in countries facing severe and increasing climate risks. Young people will bear a disproportionate burden of future climate change costs, making their ability to interpret climate information and assess changing risks important for long-run mitigation and adaptation. Yet the pathways through which young people form and revise perceptions about climate change remain uncertain. Extreme weather events are hypothesized to shape perceptions through lived experience (Akerlof et al., 2013; McDonald et al., 2015), while school-based information interventions are increasingly used to shift climate-related knowledge and attitudes (Bottin et al., 2023). Understanding whether these information and experience channels meaningfully affect young people's climate change perceptions is therefore central to designing effective climate communication and adaptation strategies.

This paper explores the updating of climate knowledge, beliefs, and attitudes in Bangladesh, one of the most climate-exposed countries in the world, facing recurrent flooding, riverbank erosion, sea level rise, and increasingly unpredictable rainfall that shape daily life. We create a three-wave panel dataset from more than 5,000 Bangladeshi students, tracked from grade 8 to 9, to answer the following questions. What factors explain variation in knowledge, beliefs, and attitudes related to climate change? Are these outcomes altered by new information and exposure to climatic shocks?

Our focus is on both an educational intervention and a severe storm event, aligning with two pathways emphasized in the broader literature that may plausibly reshape beliefs: new information and lived experience. There has been substantial interest in school-based climate-related information interventions, particularly in settings where young people face substantial future climate risks and where schools offer a scalable way to reach large cohorts of students. The most closely related study in our setting is the cluster-randomized climate change and health education intervention conducted by Kabir et al. (2015) in Bangladesh. This six-month curriculum combined student materials with weekly 45-minute teacher-led sessions and produced substantial gains in climate and health knowledge, demonstrating that sustained classroom engagement can raise climate literacy in climate-vulnerable regions. Evidence from simpler, stand-alone interventions is more mixed. Abrham et al. (2023) found no clear improvement in adolescents' understanding following a single one-hour session, although some self-reported behavioral responses changed immediately after the intervention. Shalini et al. (2025) evaluated a school-based intervention that used a child-to-child approach and flip charts to teach students ages 12–16 about climate change and its health impacts. At a six-week follow-up, students demonstrated improved climate knowledge, while changes in attitudes and practices were more limited.

Systematic reviews generally find that environmental education programs are more likely to shift attitudes when they include experiential components, local relevance, or sustained engagement (Bottin et al., 2023; Monroe et al., 2019). This distinction matters for climate adaptation policy because brief information campaigns remain appealing in resource-constrained settings. They are low-cost, scalable, and easy to integrate into schools. Bottin et al. (2023) also caution that the unusually high proportion of positive findings in the literature may partly reflect selective publication or reporting, suggesting that unsuccessful or short-lived interventions may be underrepresented. Governments and implementing organizations may therefore overestimate the extent to which one-off climate messaging can build durable climate awareness. Evidence of limited or null effects is especially policy-relevant in frontline countries where education and adaptation resources are scarce and repeatedly implementing ineffective programs carries real opportunity costs.

Complementary to planned educational interventions, a second strand of literature examines whether exposure to extreme weather events can prompt belief updating about climate risks. Several studies find that recent local shocks can increase concern in the short run, but these concerns tend to attenuate. For instance, exposure to wildfires has a small but significant effect on acknowledging climate change and supporting action, though this impact quickly fades and other event types show little effect (Visconti & Young, 2024). Similarly, flood experience can strengthen belief in climate change among those living near affected areas, but the effect declines sharply with distance and is concentrated among non-skeptics (Osberghaus & Fugger, 2022). The concern is that if lived experience generates only temporary salience, then lived experience alone may be insufficient to support durable belief updating or preparedness, and recurring hazards may be normalized rather than interpreted as signals of changing climate risk.

Our novel dataset allows us to explore how student perceptions and awareness to climate change evolves over the 14-month study period, and what student characteristics are predictive of these variables. Importantly, we pair the panel data with two sources of exogenous variation: a school-based informational intervention, implemented as part of a randomized controlled trial (RCT), and quasi-experimental variation from a major storm. These two sources of variation allow us to examine both the information and experience channels discussed above.

Schools were randomly assigned at baseline to receive a short video and pamphlet on climate science and local impacts. At midline, schools were separately randomized to receive a climate science booklet and the announcement of a follow-up quiz, along with a financial reward for high performance. Midway through the administration of the midline survey, some schools were exposed to a very large storm event. Exploiting the fact that some schools were surveyed shortly after the storm while others were surveyed just before, we estimate short-run effects of this event on climate beliefs. Together, we provide experimental evidence on belief updating in response to new information and quasi-experimental evidence on exposure to extreme rainfall and find no evidence that students update their beliefs following either the education intervention or recent storm exposure

Additionally, we examine baseline correlates of students' climate-related knowledge, attitudes, and beliefs and connect these patterns to a broader literature documenting educational gradients in climate beliefs and behavior. Education is associated with lower climate skepticism and greater climate concern (Poortinga et al., 2019), as well as stronger proenvironmental norms and intended energy-saving behavior (Weckroth & Ala-Mantila, 2022). Cross-national evidence also links education with climate knowledge and climate-policy preferences, although knowledge alone does not fully account for support for particular policies (Dechezleprêtre et al., 2025). These relationships extend beyond climate attitudes to trust in science across domains, from climate science to vaccines, although the strength and direction of these gradients can vary with political identity (Hamilton et al., 2015). Issue framing and perceptions of technological responses can further shape trust in climate science (Fairbrother, 2016; Morin-Chassé & Lachapelle, 2020).

Our setting provides a complementary perspective. Rather than comparing individuals with different levels of completed schooling, we observe variation in climate and math quiz performance among students at the same grade level. We therefore extend the evidence on educational gradients in climate beliefs by showing that climate-related attitudes vary with academic performance and science-relevant knowledge, proxied by performance on the math and climate quizzes, even when formal schooling is held constant.

The remainder of the paper proceeds as follows. Section 2 describes the data and provides an initial analysis of student climate-related outcomes and their determinants. Section 3 presents our experimental climate education study. Section 4 presents our quasi-experimental storm exposure study. Section 5 concludes.

2 Data and Preliminary Analysis

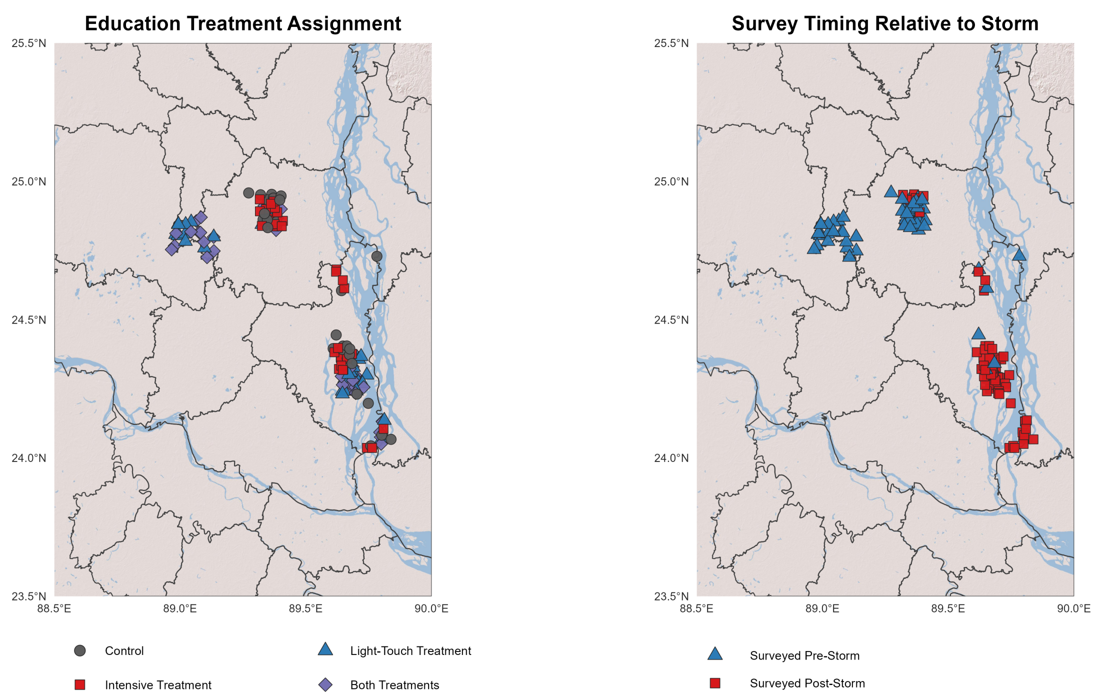
2.1 Data

The data for all analysis within this paper come from a three-wave panel survey originally collected as part of a school-level RCT, which provided the framework for school selection, timing of data collection, and variables measured.

2.1.1 Sampling and Randomization

The study was conducted in the Rajshahi division of Bangladesh.[1] Given Bangladesh's susceptibility to intense monsoon rains and flooding, districts with average rainfall above the division's median were examined across historical regional exposure to rainfall. [2] Figure 1 shows the locations of the study schools, their education treatment assignments, and their survey timing relative to the October 5, 2023 storm that will serve as our second external shock.

Figure 1: School locations and treatment assignment



Note: Points denote school locations in Bangladesh by education treatment assignment and survey timing relative to the storm. School coordinates, treatment assignments, and survey dates come from the study data. District boundaries are from GADM version 4.1, accessed using the `geodata` R package. Esri World Shaded Relief map service. *Basemap attribution:* Copyright:(c) 2014 Esri.

Schools were assigned to treatment groups at the school level in a 2×2 design that cross-randomized a light-touch baseline information component and a more intensive midline

[1]Divisions are the largest administrative region in Bangladesh. There are eight divisions in Bangladesh.

[2]Districts with long-run average monsoon-season rainfall above the Rajshahi Division median were classified as historically high exposure, while districts below the median were classified as historically low exposure.

component, yielding four treatment arms: pure control, baseline (light-touch) only, midline (intensive) only, and both components.[3]

Within each district, 70 schools were allocated across arms as follows: 17 pure control, 17 light-touch only, 18 intensive only, and 18 receiving both components. Pooling across districts yields 34, 34, 36, and 36 schools in each arm, respectively.[4] All enrolled grade 8 students in the sampled schools were eligible for the survey. This produced a baseline sample of more than 5,600 students, along with a teacher sample of roughly 560 teachers.

2.1.2 Survey Overview and Timeline

The study followed students and teachers across three survey waves from early 2023 (grade 8) to mid 2024 (grade 9), as summarized in Figure 2. We refer to these waves as baseline (BL), midline (ML), and endline (EL). The baseline survey was administered in March 2023, along with the baseline treatment, and included both student and teacher questionnaires. Six months later, during September and October 2023, the midline survey was conducted. The midline survey immediately preceded the midline treatment, which consisted of a detailed climate science booklet and notification of an upcoming test tied to performance rewards. Finally, the endline survey took place in May 2024 and involved in-person questionnaires, as well as climate and math quizzes. The analysis sample is restricted to students observed at baseline, when demographic information was collected.[5]

Four eighth grade teachers available on the survey day were interviewed at both baseline and midline. Teachers in these schools rotate frequently across subjects and grade levels, and students do not have a single assigned teacher. For this reason, teacher variables in the analysis are averaged to the school level (generating school-level variables like average

[3] Although we refer to the second intervention as "intensive" because it was more intensive than the first intervention, it remained relatively light-touch.

[4] Two schools did not remain in the sample: one had too few students in the eligible grade, while the other school discontinued the grade being observed in the study.

[5] Student surveys were administered in classroom sessions using paper questionnaires that were later digitized, with baseline students linked across waves using pre-generated barcode identifiers affixed to survey booklets.

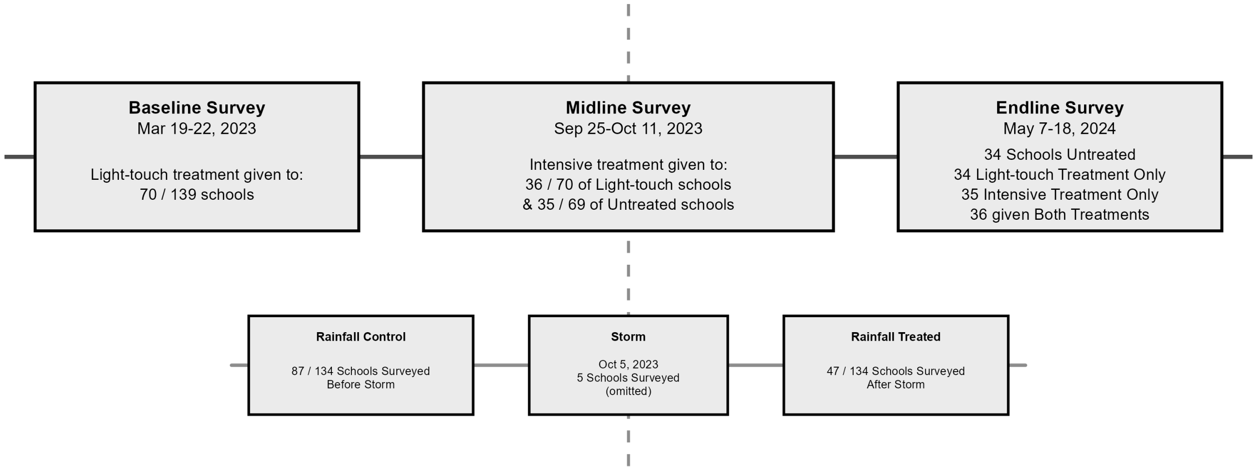


Figure 2: Study Timeline

teacher climate knowledge and average teacher age, for example).[6]

2.1.3 Primary Outcome Variables

Our primary outcomes consist of five measures: climate and math quiz performance and three composite indices capturing climate skepticism, climate-related anxiety, and problem-solving engagement. The climate quiz is intended to proxy climate knowledge, while the math quiz provides a rough proxy for STEM performance or aptitude, though it may also reflect test-taking effort or ability. Math quiz performance was originally included as an outcome because we hypothesized that increased knowledge about or interest in climate change might induce greater student effort in STEM classes.

All non-quiz outcomes are formed from either Likert or yes/no responses and follow a consistent index construction procedure. Individual responses are coded so that higher values correspond to greater skepticism, anxiety, or problem-solving engagement. Each item is standardized to have mean zero and standard deviation one using the baseline pure-control group mean and standard deviation, and the standardized items are then averaged to form each index. Quiz-based outcomes, which record the share of correct answers, are standardized

[6] Teacher surveys were administered as tablet-based interviews (SurveyCTO) at baseline and midline; teachers were not surveyed at endline.

using the total quiz score. Several additional behavioral items were collected only at later waves (e.g., discussions with friends; endline donation and migration intentions); we analyze these separately rather than folding them into the core indices.

The specific components of each index are listed in Appendix Table A1. The index components were pre-determined in our registered pre-analysis plan.[7] The climate skepticism index aggregates items capturing doubts about the human causes of climate change, media exaggeration of climate change, and skepticism about projected future consequences. The climate anxiety index combines items measuring worry, fear, and stress related to current and future climate conditions. The final index, which we refer to as problem-solving engagement, captures students' willingness to address climate change and their confidence in collective responses. Its components reflect a trust in collective action, in the ability of scientists and others to solve climate-related problems, and a willingness to join in the effort. Across the indices, higher values indicate greater skepticism, anxiety, and engagement, respectively. At endline, students made a real-stakes decision about whether and how much of their survey reward to donate to environmental causes.

2.1.4 Independent Variables

The independent variables include student gender, age, and a proxy for household wealth, along with indicators for residence in the historically high-exposure district and whether the student's mother and father completed secondary education. [8] Teacher characteristics include school-level averages of climate knowledge, skepticism, anxiety, age, gender composition, and the share with a Master's degree. Extended specifications also condition on each student's baseline climate and math quiz scores.

The analysis sample includes only students who were present at baseline, when demographic information was collected. For continuous independent variables, missing values are

[7] The pre-analysis plan can be found under the AEA RCT Registry trial AEARCTR-0011254.

[8] The household wealth index is based on asset ownership on a set of goods (e.g., phones and bicycles), constructed using the same standardization procedure as the outcome indices.

replaced with school-level means, and categorical variables with missing values receive an additional category for missing values. This preserves observations that would otherwise be lost through listwise deletion.

2.1.5 Summary Statistics

Table 1 presents summary statistics for our outcome variables and baseline covariates, comparing across treatment arms. The first column reports the mean for the control group, while the next three columns provide the differences between each of the treatment groups and the control. For binary outcomes, the outcome represents a proportion. Most baseline differences are modest, although several covariates differ significantly across individual treatment-control comparisons. Most notably, math scores are significantly higher in the baseline treatment compared to the control arm. As described in section 3, we explicitly account for these imbalances by using a double lasso procedure to select control variables for our regressions.

2.1.6 Rainfall

We use the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) dataset to characterize rainfall at the study schools (Funk et al., 2015). CHIRPS combines satellite estimates with available rain-gauge observations to produce daily precipitation estimates at a spatial resolution of $0.05° \times 0.05°$. Each school is assigned the rainfall series of the CHIRPS grid cell containing its coordinates. Consequently, schools located within the same grid cell share the same rainfall series; the 140 study schools map onto 63 unique grid cells.

The October 5, 2023 storm was the most severe rainfall event of the year at every study school, and daily rainfall exceeded 100 mm in all 63 grid cells. Appendix Figure A2 documents rainfall on this day was substantially heavier than rainfall on the largest single-day observed at the same schools during the preceding five years. Because the event was severe throughout the study area and cross-school variation in storm intensity was limited, the

Table 1: Summary statistics, by RCT treatment group

	Mean	Mean(Group) - Mean(Control)		
	Control	Baseline Treatment	Midline Treatment	Both
Outcomes				
CC quiz (BL, standardized)	-0.061	0.158 (0.102)	0.012 (0.105)	0.078 (0.122)
CC skepticism (BL, standardized)	0.037	-0.096* (0.054)	-0.022 (0.043)	-0.020 (0.042)
CC engagement (BL, standardized)	-0.022	0.068 (0.047)	0.011 (0.044)	-0.011 (0.043)
CC anxiety (BL, standardized)	-0.028	0.048 (0.050)	0.045 (0.040)	0.022 (0.050)
Math score (BL, standardized)	-0.166	0.431*** (0.155)	0.019 (0.159)	0.224 (0.141)
Donation amount (EL)	19.763	2.994 (2.476)	1.252 (2.543)	2.015 (2.616)
Baseline covariates				
Female	0.674	-0.044 (0.067)	-0.128* (0.076)	-0.077 (0.071)
High climate exposure	0.472	0.112 (0.130)	-0.022 (0.132)	0.092 (0.129)
Student age	13.799	-0.062 (0.080)	-0.061 (0.081)	-0.105 (0.075)
Mother secondary education	0.616	0.028 (0.042)	0.052 (0.041)	0.034 (0.039)
Father secondary education	0.588	0.063 (0.041)	0.062 (0.042)	0.031 (0.038)
Asset index, (mean zero)	-0.122	0.234** (0.115)	0.123 (0.141)	0.139 (0.113)
Teacher climate knowledge	0.839	-0.006 (0.015)	-0.022* (0.013)	-0.015 (0.015)
Teacher climate anxiety	0.574	-0.041* (0.024)	-0.046** (0.022)	-0.026 (0.019)
Teacher is female	0.301	-0.116** (0.052)	-0.014 (0.061)	-0.013 (0.058)
Teacher age	42.974	-0.878 (0.941)	0.343 (0.931)	1.090 (0.961)
Teacher has M.A. degree	0.683	-0.057 (0.070)	-0.124* (0.070)	-0.275*** (0.064)
Observations	1530	1421	1369	1330

Notes: Each difference column reports Mean(Group) - Mean(Control), estimated from separate regressions of the outcome on a group indicator using only Control and the corresponding treatment group observations. Standard errors are clustered at the school level. Significance levels: $*p < 0.10, **p < 0.05, ***p < 0.01$.

CHIRPS measures are used descriptively rather than as the primary source of identifying variation. Our primary analysis instead relies on survey timing relative to the storm, as described in Section 4.

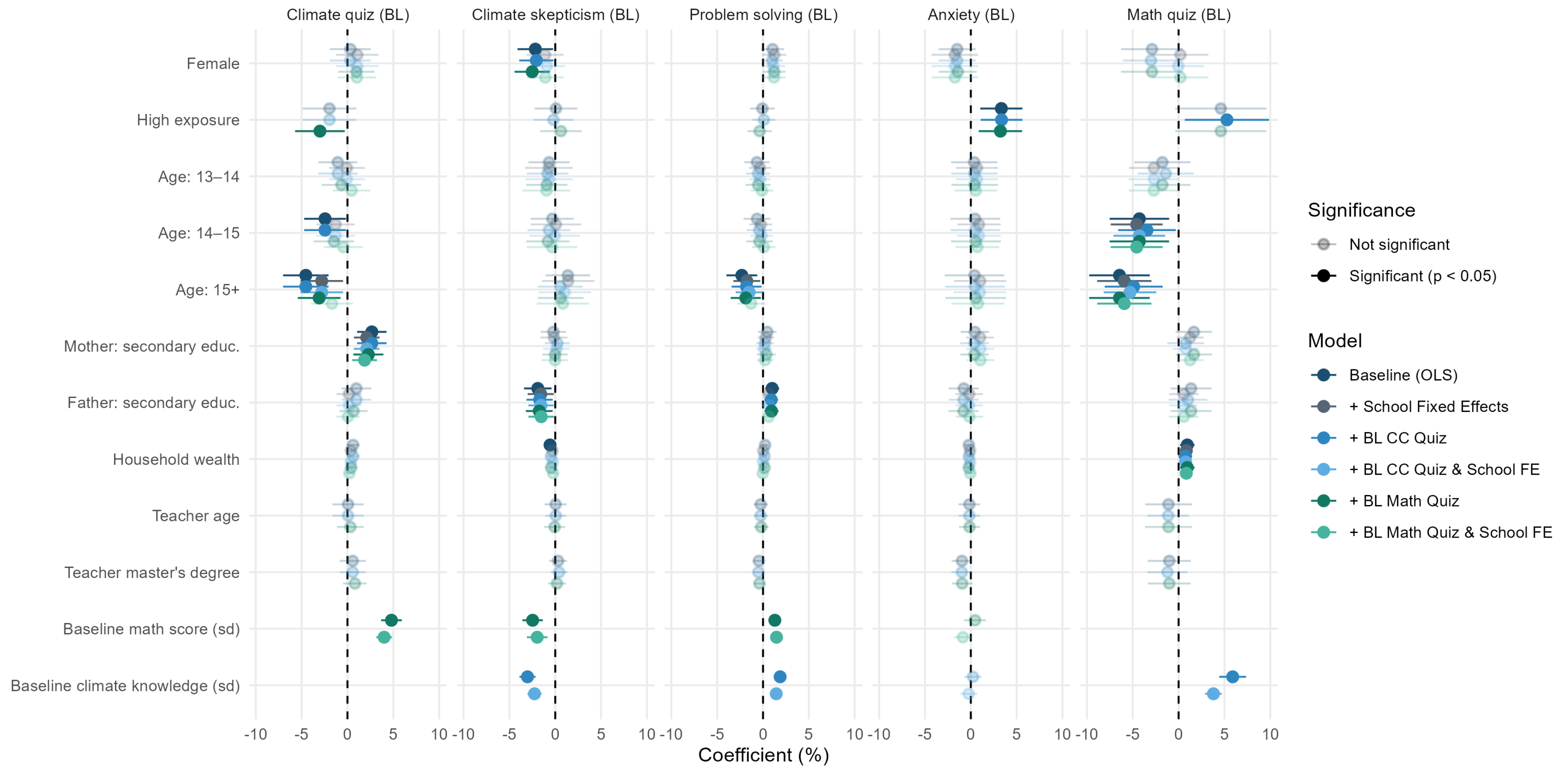
2.2 Preliminary Analysis

Appendix Figure A1 summarizes baseline responses for the full sample and across key student subgroups. Two patterns are especially relevant for our analysis. First, students enter the study with relatively high recognition of climate change, but with important gaps in certainty, personal experience, and perceived agency. In the full sample, only 16 percent of students doubt the human-driven nature of climate change, but 30 percent believe climate change is exaggerated in the media and 24 percent doubt the likelihood of projected climate consequences by 2050. This suggests that basic recognition of anthropogenic climate change does not necessarily imply confidence in predicted future impacts or acceptance of climate projections. Students also vary in their reports of observing environmental change and experiencing it directly: 45 percent report noticing climate-related changes in places important to them, while only 21 percent report knowing someone directly affected by climate change and 20 percent report being directly affected themselves. This distinction is central to our later rainfall analysis, as it suggests that exposure to environmental change may not automatically translate into personal attribution or belief updating.

There are also differences in climate understanding across groups. Differences by demographic and socioeconomic characteristics are present, but the clearest gradients appear by age-for-grade and baseline quiz performance. Students in the oldest age-for-grade cohort, likely those who have had to repeat grades, perform worse than those in the youngest cohort on both the climate quiz, 65 percent versus 71 percent, and the math quiz, 41 percent versus 49 percent. Students in the higher initial math group score 72 percent on the climate quiz, compared with 64 percent in the lower group, and are more likely to plan to donate. These subgroup patterns motivate the regression analysis below, which asks whether the same gradients remain after conditioning on other student, household, teacher, and school-level characteristics.

Figure 3 provides a visual summary of the regressions of baseline outcomes onto demographic and teacher information. We also estimate specifications that add baseline math and climate quiz scores and school fixed effects, allowing us to compare students within the same school. The regression results reinforce the aforementioned descriptive patterns after controlling for other observables. The strongest and most consistent predictors are students' own baseline quiz scores. A one-standard-deviation increase in baseline math-quiz performance is associated with approximately a five-percentage-point increase in baseline climate-quiz accuracy and a 0.04–0.05 standard-deviation reduction in the skepticism index. Higher math scores also predict greater problem-solving engagement, which largely reflects confidence in collective action and science. Baseline climate-quiz performance shows similar relationships with climate-related beliefs and attitudes. The climate quiz relationship is partly expected because several items directly test concepts that overlap with skepticism, including human emissions, fossil fuels, deforestation, agriculture, and projected impacts. The association with the math quiz is particularly notable because that quiz contains no climate-related content.

Figure 3: OLS coefficients of baseline covariates on baseline outcomes



Note: Each column of coefficients corresponds to its own regression.

Other demographic differences are generally more modest after adding the full set of controls. Gender gaps are small across most outcomes, although female students tend to be less skeptical in most specifications. Regional climate exposure has mixed associations: students in higher-exposure areas score lower on the climate quiz, conditional on math quiz performance, but report higher climate anxiety. Parental education is associated with higher climate knowledge and lower skepticism, though these relationships are clearest for the climate quiz. Teacher characteristics are not consistent predictors of student climate outcomes.

These academic gradients are also closely linked to household wealth and age-for-grade, connecting differences in climate understanding to broader patterns of educational inequality. Students from more disadvantaged households, or students already falling behind academically, may be less equipped to interpret climate information and translate concern into adaptive responses. Evidence from disaster-prone communities suggests that formal education can strengthen adaptive capacity and risk-reduction knowledge, although access to these opportunities remains unequal (Wamsler et al., 2012). Our results show that, even within a single level of formal schooling, socioeconomic and academic divides remain closely tied to climate understanding.

3 Climate Education RCT

Moving on to explore external shocks, we first explore the extent to which information can shift climate knowledge, beliefs, and attitudes. For this, we rely on an RCT that generated exogenous variation in information through two interventions.

3.1 Treatments

The baseline treatment was a light-touch intervention consisting of a short video that introduced core concepts of climate science, including greenhouse gases, temperature rise, and local environmental impacts. The video focused specifically on how Bangladesh is likely to be affected by climate change and was paired with a simple pamphlet reiterating these ideas in accessible language. The midline treatment, implemented roughly six months later, provided students with a detailed climate-science booklet emphasizing scientific reasoning, adaptation strategies, mitigation strategies, and locally relevant climate challenges. Students were informed that they would later take a quiz on the booklet and that high performers would receive small prizes. This was intended to encourage careful reading and engagement with the material.

3.2 Empirical Strategy

The RCT provides exogenous variation in climate education exposure by randomly assigning schools to the treatment arms described above. Our primary estimation approach follows an ANCOVA specification that conditions on baseline outcomes and includes a set of covariates selected via post–double-selection lasso (BELLONI et al., 2014). Standard errors are clustered at the school level, the level of randomization, and all models include district fixed effects.

The main estimating equation is:

$$Y_{it} = \beta_{t1}\text{Baseline Treatment}_j + \beta_{t2}\text{Midline Treatment}_j + \gamma_{t1}Y_{i0} + \delta_{t2}X_{ij0} + \mu_d + \epsilon_{ijt} \quad (1)$$

Where, Y_{it} denotes the outcome for student i at time (i.e., survey wave) t. The variables Baseline Treatment$_j$ and Midline Treatment$_j$ are dummy variables indicating whether school j received the relevant intervention. The term Y_{i0} is the student's baseline value of the same outcome, and X_{ij0} represents the baseline covariates selected through the double lasso procedure. District fixed effects are captured by μ_d.

The complete pool of baseline covariates possible for vector X_{ij0} includes gender, age categories, parental education, household wealth, and several school-level teacher characteristics such as average climate knowledge, skepticism, anxiety, gender composition, age, and the share of teachers with a Master's degree. In extended specifications, we also include baseline climate and math quiz scores. This approach systematically reduces overspecification and preserves precision by identifying only the covariates that predict either treatment assignment or the outcome. Because the survey data include three waves and two interventions, we also estimate short-term specifications for baseline-to-midline and midline-to-endline intervals to capture immediate responses to each intervention:

$$Y_{it} = \beta_{t1}\text{Treatment}_j + \gamma_{t1}Y_{i0} + \delta_{t2}X_{ij,t-1} + \mu_d + \epsilon_{ijt} \tag{2}$$

The short-term specification allows us to estimate treatment effects separately for the first six months (baseline to midline) and the subsequent eight months (midline to endline), which is important because nearly fourteen months pass between baseline and endline, creating substantial opportunity for effects to attenuate.

Missing outcomes arise for two reasons: students may attrit from the sample in later waves, or students may remain in the sample but skip specific questions. We follow the protocol in the original analysis plan to diagnose selective non-response. For each outcome Y_{it}, we estimate the probability of response based on observable characteristics using the following model:

$$\begin{aligned} N^Y_{ijt} = {} & \alpha_1\text{Baseline Treatment}_j + \alpha_2\text{Midline Treatment}_j \\ & + \alpha_3\text{Baseline Treatment}_j \times \text{Midline Treatment}_j + \alpha_4 Y_{i0} + \alpha_5 X_{ij0} + \mu_d + \epsilon_{ijt} \end{aligned} \tag{3}$$

A non-missing value for outcome Y is denoted by $N^Y_{ijt} = 1$. All other terms are as specified above. We test whether any of the treatment coefficients are individually or jointly significant at the 10 percent level to determine selective response. When selective non-response is detected, we construct inverse probability weights following the pre-analysis plan.

To construct the inverse probability weights, we estimate an expanded probit version of the attrition equation above, adding interactions between each treatment dummy and both Y_{i0} and X_{ij0}. The predicted probability of being observed is then used to compute:

$$w_{ijt} = \frac{1}{\widehat{P}\left(N_{ijt}^{Y} = 1\right)}. \tag{4}$$

Each outcome is weighted by its corresponding w_{ijt} in all ANCOVA regressions. This re-weights the observed sample to represent the baseline sample, correcting for differential attrition across treatment arms. The ANCOVA specification, covariate selection, attrition tests, and construction of inverse probability weights follow the same structure across all outcomes and both survey intervals. This ensures comparability of estimates and allows us to evaluate both short-term and medium-term updating of climate knowledge, beliefs, anxiety, and engagement.

3.3 Results

Table 2 reports the endline treatment-effect estimates in two panels. Panel A presents the primary ANCOVA specification, while Panel B additionally controls for baseline climate- and math-quiz performance. Across both specifications, neither the baseline nor the midline treatment produces a statistically detectable change in climate-quiz performance, skepticism, anxiety, problem-solving engagement, math-quiz performance, donation amount, or the probability of donating. Additional short-term specifications measuring the impacts of the Baseline Treatment at midline and the Midline Treatment at Endline similarly show no statistically detectable impacts approximately six months after each intervention (Appendix Table A2). Neither the baseline video and pamphlet intervention nor the midline booklet with incentivized studying produced detectable changes in climate knowledge, climate skepticism, or climate anxiety. Short-term effects measured at midline or at endline, also show no meaningful impacts approximately six months after each intervention (See Table A2).

Table 2: Education Treatment Results

Panel A: Primary Specification							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0255	0.0098	0.0179	-0.0135	-0.1283	2.162	0.0226
	(0.0731)	(0.0298)	(0.0317)	(0.0286)	(0.0889)	(1.793)	(0.0374)
Midline Treatment	-0.0198	-0.0029	0.0150	0.0078	-0.0708	1.752	-0.0163
	(0.0723)	(0.0290)	(0.0335)	(0.0299)	(0.0811)	(1.845)	(0.0385)

Panel B: Primary Specification Controlling for BL Math and BL Climate							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0558	0.0285	0.0063	-0.0313	-0.1356	1.769	0.0217
	(0.0703)	(0.0289)	(0.0352)	(0.0281)	(0.0883)	(1.755)	(0.0391)
Midline Treatment	-0.0129	0.0024	0.0216	0.0098	-0.0706	1.964	0.0021
	(0.0692)	(0.0303)	(0.0337)	(0.0290)	(0.0814)	(1.881)	(0.0401)
Baseline Climate Knowledge	0.1373***	-0.0527***	0.0173	0.0426***	0.1096***	-0.3598	-0.0036
	(0.0218)	(0.0093)	(0.0127)	(0.0109)	(0.0238)	(0.4549)	(0.0109)
Baseline Math Score	0.1172***	-0.0302***	0.0299*	0.0383***	0.1710***	1.466**	0.0212*
	(0.0271)	(0.0109)	(0.0159)	(0.0125)	(0.0388)	(0.6176)	(0.0127)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes	No	No
Observations	3,732	3,690	3,721	3,705	3,732	3,732	3,732
Mean of DV	0.023	-0.007	-0.003	0.019	0.015	21.275	0.726

Notes: The summary stats provided correspond to Panel A: Primary Specification
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Although the treatment coefficients are statistically insignificant, Panel B also demonstrates substantial persistence in academic and climate-related outcomes. Higher baseline climate- and math-quiz performance predict higher endline climate-quiz performance, lower climate skepticism, stronger stated problem-solving engagement, and greater revealed engagement, measured through endline donation behavior. The relationship between baseline climate-quiz performance and skepticism may be partly mechanical because several quiz items concern the anthropogenic nature of climate change and expected climate impacts, which overlap with components of the skepticism index. However, the corresponding relationship with math-quiz performance cannot be explained by direct climate content. These relationships are descriptive and may reflect academic skill, survey effort, motivation, or other student characteristics rather than causal effects of knowledge. At the same time, they are consistent with beliefs and engagement being shaped by knowledge and skills accumulated over a longer period, which may make them less responsive to a single exposure to new information. Short-term interventions may therefore be insufficient to produce detectable changes in outcomes that have formed and persisted over much longer periods.

Heterogeneity

Although full-sample treatment effects are null, these average effects may be masking significant effects for specific groups. As specified in our pre-analysis plan, we examine heterogeneity by gender and district. Panel A of Appendix Table A3 reports results for female students, panel B reports results for male students, and panel C reports the gender difference in the estimated treatment effects. There is no evidence of significant heterogeneity across genders, nor is there strong evidence for significant effects in either of the groups. One exception is that the baseline treatment significantly increased donation amounts for male students ($p < 0.1$), though this coefficient is not significantly different from the positive coefficient for female students.

We next examine whether treatment effects vary across the historically high- and low-exposure districts. Panel A of Appendix Table A4 reports results for the high-exposure district, Panel B reports results for the low-exposure district, and Panel C reports the difference between them. Two cross-district differences are statistically significant: the baseline-treatment coefficient for climate skepticism is more positive in the high-exposure district, while the midline-treatment coefficient for problem-solving engagement is more negative. However, neither underlying treatment coefficient is itself statistically significant within either district. Only one subgroup-specific treatment coefficient is statistically significant. The baseline-treatment for donation amount shows a 3.7 BDT increase in endline donation for students in the high-exposure district who received the baseline treatment. However, this may be a spurious effect given the large number of coefficients being tested and the fact that the more intensive and more recently experienced midline treatment shows no effect.

Interpreting these exposure-related differences requires caution. First, although we explicitly stratified districts by climate risk and selected one high exposure and one low exposure district, climate risk is not the only characteristic that distinguishes these two districts. There are likely to be additional unobserved differences between these districts beyond historical rainfall exposure, making it difficult to attribute any differences we find here to variation in climate risk. Second, interpretation is further complicated by the fact that the study period coincided with large-scale flooding in the entire study area. As a result, by the endline survey, both districts had been recently exposed to extreme rainfall, which could make any historical rainfall differences irrelevant. These issues motivate the next step of the analysis, which merges high-resolution daily rainfall data to schools in order to isolate the role of extreme precipitation shocks during the RCT period.

4 Storm Exposure

Survey implementation for the climate education RCT proceeded region by region. A major storm occurred across the study area in Rajshahi Division on October 5, 2023, after 87 schools had completed the midline survey but before another 47 were surveyed. Five schools surveyed on the date of the storm are excluded from the principal analysis. We focus exclusively on the baseline-to-midline (BL–ML) window as this window provides a clean comparison between schools recently exposed to the storm and those not yet exposed. Conversely, the midline-to-endline (ML–EL) period lacks a true control group because all schools eventually experienced the storm.

As discussed in Section 2, CHIRPS indicates that the event produced unusually high rainfall throughout the study area but provides limited cross-school variation in storm intensity. The analysis therefore relies on the extensive margin of recent rainfall exposure, comparing students surveyed shortly after the storm with those surveyed beforehand.

4.1 Rainfall Shock Specification

We define a dummy variable, RainShock_j, set to equal one for schools that were exposed to the storm prior to taking the midline survey. Using this as our main independent variable of interest, we estimate the following regression, which parallels the short-term ANCOVA used in the RCT analysis:

$$Y_{i1} = \beta_1 \text{RainShock}_j + \gamma Y_{i0} + \delta X_{ij0} + \mu_d + \epsilon_{ij1}. \tag{5}$$

Y_{i1} is the midline outcome of interest, Y_{i0} is the baseline outcome, and X_{ij0} is the vector of baseline covariates selected via post–double-selection lasso. District fixed effects are included, and standard errors are clustered at the school level. Schools taking the exam on the day of the storm are excluded from the sample. In this regression, β_1 captures the average effect of the storm within approximately one week of the event.

Table 3 reports the full-sample estimates for this specification. We find no evidence that storm exposure generated changes in knowledge, beliefs, or attitudes. All coefficients are statistically insignificant and most of them are small in magnitude, representing effect sizes of less than 2% of a standard deviation (with the exception of the climate knowledge regression).

Table 3: Impacts of Rainfall Shock

Rainfall Impacts Baseline–Midline					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Exposed to Rainfall Shock	0.1438	0.0048	0.0078	0.0136	0.0052
	(0.1208)	(0.0430)	(0.0447)	(0.0436)	(0.1127)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes
Observations	3,865	3,810	3,843	3,801	3,865
Mean of DV	0.063	-0.018	-0.002	0.022	0.074

Notes: Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

It is important to note that, although the storm was an unexpected event, it did not split the sample into two balanced groups (unlike the education RCT, which was explicitly randomized). In fact, Table A6 highlights several significant differences across the two rainfall groups: those exposed to the storm before midline were younger, had lower maternal education, household assets, and female-teacher representation. They also had higher climate anxiety and higher math scores at baseline. One particularly large and potentially quite consequential imbalance is that almost all schools exposed to the storm before midline were in the high exposure district. While the regressions in Table 3 do address these imbalances by controlling for baseline covariates, including district fixed effects, the high correlation between the high-exposure dummy and rainfall shock variable could make it difficult to uncover precisely estimated storm effects even if storm effects do exist. We therefore also estimate this same regression restricting to high exposure districts within which there is adequate variation in rainfall shock exposure (unlike in low exposure districts, where almost no schools were exposed to the storm by midline). Results of the high exposure portion are reported in panel B of Appendix Table A5, revealing statistically insignificant coefficients for all outcomes.

Lastly, we jointly estimate the effects of the education-treatment, rainfall-timing, and interaction terms. The specification can only be estimated on the set of outcomes available at midline, but should identify any additive effects of the information intervention and experiential exposure. The interaction term tests whether the effect of the baseline education treatment differs between students surveyed after the storm and those surveyed beforehand. Appendix Table A7 presents the estimates: the main effects and interaction coefficients are statistically insignificant across all five outcomes.

5 Conclusion

We examine the factors associated with climate knowledge, skepticism, anxiety, and problem-solving engagement among Bangladeshi adolescents and assess whether these outcomes respond to climate education or recent exposure to extreme rainfall.

The descriptive patterns add a useful perspective to prior work documenting differences in climate beliefs, skepticism, trust in climate science, and support for climate policies across levels of educational attainment (Dechezleprêtre et al., 2025; Fairbrother, 2016; Hamilton et al., 2015; Poortinga et al., 2019; Weckroth & Ala-Mantila, 2022). Our setting differs because we measure variation in knowledge and cognition across students observed within the same grade level. Even among students with the same level of formal schooling, climate knowledge, skepticism, and problem-solving engagement are strongly associated with baseline quiz performance measured 14 months earlier. This pattern provides suggestive evidence that the educational gradients documented elsewhere may partly reflect differences in the academic skills and climate-relevant knowledge that students accumulate alongside formal schooling, rather than differences in schooling level alone. However, students who performed well on both the math and climate portions of the survey may also have been more motivated or attentive throughout the survey, so these descriptive relationships should be interpreted cautiously.

We then test two pathways for belief updating: direct exposure to new climate information and exposure to a major rainfall shock. Across both empirical studies, we find little evidence of belief updating. These results show a persistence in measured climate outcomes among adolescents despite brief information campaigns and recent exposure to a major climate-related shock. The null effects from the randomized interventions are consistent with prior work showing that short, stand-alone interventions often produce limited or short-lived changes in attitudes or behavior (Abrham et al., 2023; Ramadani et al., 2023; Shalini et al., 2025). While our results do not provide direct evidence on more intensive climate education programs, they may help explain why light-touch programs often fall short. Brief exposure to information may be insufficient when students differ in the prior knowledge and academic skills they bring to that information. However, longer-lasting and more intensive programs have produced stronger effects in very similar settings (Kabir et al., 2015). Preparing young people for a changing climate may require greater investments that connect climate curricula with the foundational academic skills needed to interpret risk, trust evidence, and make informed adaptation decisions.

The storm-exposure analysis provides complementary evidence on the impacts of extreme weather experience. Exposure to climate events is often hypothesized to alter one's perceptions by making climate change more tangible and revealing latent risk, but our results show that lived experience alone may not be enough to produce meaningful updating. Students surveyed shortly after a major rainfall event show little evidence of a shift in skepticism, anxiety, or engagement, implying that experiential learning from climate shocks does not universally translate into updated climate beliefs among adolescents. This aligns with prior evidence that local climate shocks produce limited or short-lived belief change (Osberghaus & Fugger, 2022; Visconti & Young, 2024). When climate shocks do shift attitudes in the near term, recurring hazards may be normalized rather than interpreted as signals of changing climate risk. Taken together, our study demonstrates a limited path to sustained shifts in climate beliefs through natural exposure to storms or through light-touch education interventions.

Declaration of generative AI and AI-assisted

During the preparation of this work, the authors used OpenAI's ChatGPT to assist with language editing, manuscript organization, and consistency review. Following use of this tool, the authors reviewed and edited all content as needed, independently verified the manuscript's substantive claims and references, and take full responsibility for the content of the publication.

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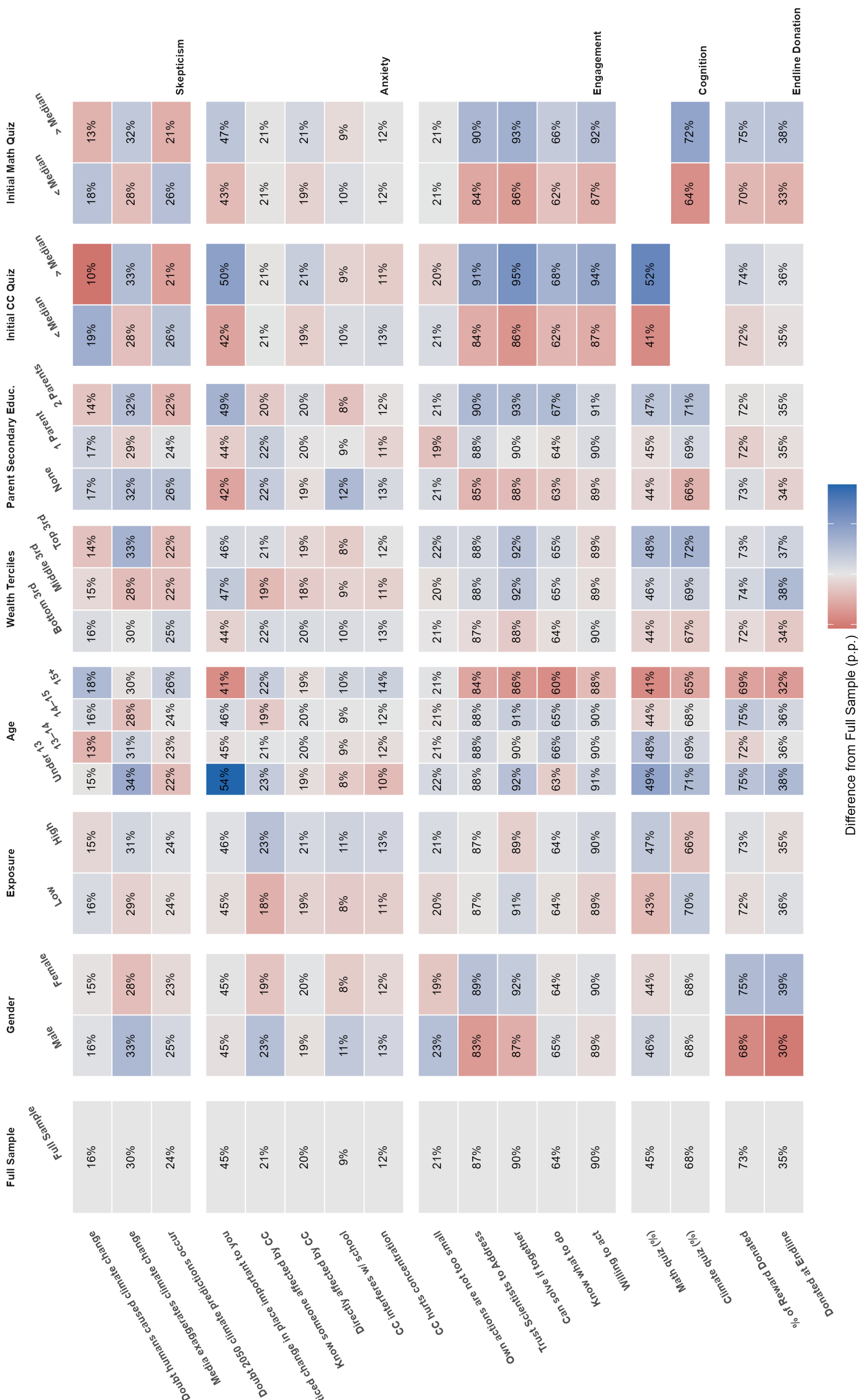
Appendix: Additional Tables and Figures

Figure A1: Baseline heatmap of index components and observables by grouping variables



Note: Several variables shown are derived from Likert-scale items recoded into binary indicators. Responses of "Often" or "Almost always" are coded as 1, and "Never," "Rarely," or "Sometimes" are coded as 0. For items using a 1–5 agreement scale, "Strongly agree" or "Agree" are coded as 1 for positively worded statements, while negatively worded items are reverse-coded so that disagreement is coded as 1. Climate-skepticism and media-exaggeration items are similarly collapsed so that agreement or high-skepticism responses are coded as 1 and all other substantive responses as 0.

Table A1: Survey items used to construct each index

Item	Full question	Response options
Skepticism		
S1	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! Climate change and its consequences are being exaggerated in the media.	Not at all true; A little true; Somewhat true; Mostly true; Completely true
S2	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! I doubt that climate change is caused by human emissions.	Not at all true; A little true; Somewhat true; Mostly true; Completely true
S3	In your opinion, what is the likelihood that the outcomes that are described above (warming climate, Bangladesh, and forced displacement) will happen by the year 2050? (Scale item; no additional statement text)	0–9 scale (anchors: Not at all likely Extremely Likely)
Anxiety		
A1	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! Thinking about climate change makes it difficult for me to concentrate.	Never; Rarely; Sometimes; Often; Almost always
A2(BL)	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! My concerns about climate change interfere with my ability to get work or school assignments done.	Never; Rarely; Sometimes; Often; Almost always
A2(ML,EL)	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! My family's security (e.g., economic, social, physical security) will be threatened.	Never; Rarely; Sometimes; Often; Almost always
A3	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! I have been directly affected by climate change.	Never; Rarely; Sometimes; Often; Almost always
A4	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! I know someone who has been directly affected by climate change.	Never; Rarely; Sometimes; Often; Almost always
A5	To what extent do you agree with each of the following statements [Select one option per row] Be honest – there are no right or wrong answers! I have noticed a change in a place that is important to me due to climate change.	Never; Rarely; Sometimes; Often; Almost always
NA		
E1	To what extent do you agree with each of the following statements [Select one option per row]. Be honest – there are no right or wrong answers! I am willing to take action to help solve problems caused by climate change.	Strongly agree; Agree; Neither agree nor disagree; Disagree; Strongly disagree
E2	To what extent do you agree with each of the following statements [Select one option per row]. Be honest – there are no right or wrong answers! I know what to do to help solve problems caused by climate change.	Strongly agree; Agree; Neither agree nor disagree; Disagree; Strongly disagree
E3	To what extent do you agree with each of the following statements [Select one option per row]. Be honest – there are no right or wrong answers! If everyone works together, we can solve problems caused by climate change.	Strongly agree; Agree; Neither agree nor disagree; Disagree; Strongly disagree
E4	To what extent do you agree with each of the following statements [Select one option per row]. Be honest – there are no right or wrong answers! I believe that scientists will be able to find ways to solve problems caused by climate change.	Strongly agree; Agree; Neither agree nor disagree; Disagree; Strongly disagree
E5	To what extent do you agree with each of the following statements [Select one option per row]. Be honest – there are no right or wrong answers! The actions I can take are too small to help solve problems caused by climate change. (Reversed)	Strongly disagree; Disagree; Neither agree nor disagree; Agree; Strongly agree

Notes: All items are identical across waves except Anxiety Item A2, shown separately as A2(BL) and A2(ML,EL). Due to a translation error in S2, students were instead asked to respond to the statement "I think that climate change is caused by human emissions." We therefore reverse coded the responses to this question such that those who disagreed with this statement were coded as agreeing with the statement "I doubt that climate change is caused by human emissions."

Table A2: Education RCT: Short Term Impacts

Panel A: Short-Term Impacts BL–ML							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score		
Variables							
Baseline Treatment	0.0132	0.0242	-0.0317	-0.0018	-0.0703		
	(0.0712)	(0.0307)	(0.0293)	(0.0304)	(0.0769)		
Panel B: Short-Term Impacts ML–EL							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Midline Treatment	0.0016	-0.0031	0.0075	0.0141	-0.0641	1.681	-0.0028
	(0.0712)	(0.0290)	(0.0348)	(0.0327)	(0.0784)	(1.820)	(0.0399)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes	No	No
Observations	2,961	2,928	2,939	2,897	2,961	3,732	3,732
Mean of DV	0.047	-0.009	0	0.033	0.039	21.275	0.726

Notes: The summary stats provided correspond to Panel B: Short-Term Impacts ML–EL
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Education RCT Heterogeneity by Gender

Panel A: Female Subsample							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0318	0.0139	0.0297	-0.0220	-0.1264	1.066	0.0264
	(0.0833)	(0.0363)	(0.0403)	(0.0345)	(0.1019)	(2.430)	(0.0434)
Midline Treatment	-0.0511	0.0272	0.0463	0.0160	-0.0503	2.162	-0.0184
	(0.0897)	(0.0346)	(0.0437)	(0.0374)	(0.1017)	(2.499)	(0.0453)
Panel B: Male Subsample							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0174	-0.0028	0.0043	0.0052	-0.1123	3.358*	0.0194
	(0.1057)	(0.0465)	(0.0525)	(0.0442)	(0.1208)	(1.758)	(0.0480)
Midline Treatment	-0.0004	-0.0390	-0.0415	-0.0117	-0.0904	0.8125	-0.0166
	(0.1030)	(0.0513)	(0.0533)	(0.0451)	(0.1101)	(1.817)	(0.0489)
Panel C: Difference (Female - Male)							
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment Difference	-0.0144	0.0167	0.0254	-0.0273	-0.0142	-2.293	0.0070
	(0.1168)	(0.0562)	(0.0633)	(0.0524)	(0.1315)	(2.595)	(0.0514)
Midline Treatment Difference	-0.0507	0.0662	0.0878	0.0277	0.0401	1.349	-0.0017
	(0.1231)	(0.0585)	(0.0667)	(0.0560)	(0.1356)	(2.655)	(0.0536)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes	No	No
Observations	3,716	3,675	3,705	3,692	3,716	3,716	3,716
Mean of DV	0.027	-0.008	-0.003	0.021	0.02	21.284	0.727

Notes: The summary stats provided correspond to Panel C: Difference (Female - Male)
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Education RCT Heterogeneity by Regional Exposure

Panel A: High Exposure Subsample

Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0132	0.0693	0.0301	-0.0294	-0.0548	3.773*	0.0738
	(0.1078)	(0.0438)	(0.0466)	(0.0398)	(0.1300)	(2.199)	(0.0474)
Midline Treatment	0.0009	0.0025	-2.1×10^{-5}	-0.0571	-0.0918	-0.3975	0.0107
	(0.1180)	(0.0442)	(0.0487)	(0.0405)	(0.1229)	(2.489)	(0.0559)

Panel B: Low Exposure Subsample

Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment	-0.0195	-0.0609	0.0086	0.0113	-0.1981	0.9810	-0.0028
	(0.0945)	(0.0385)	(0.0456)	(0.0405)	(0.1204)	(2.905)	(0.0573)
Midline Treatment	-0.0456	-0.0115	0.0224	0.0585	-0.0677	3.475	-0.0225
	(0.0831)	(0.0366)	(0.0458)	(0.0402)	(0.1071)	(2.634)	(0.0510)

Panel C: Difference (High Exposure - Low Exposure)

Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score	Donation Amount (BDT)	Chooses to Donate
Variables							
Baseline Treatment Difference	0.0064	0.1302**	0.0215	-0.0407	0.1433	2.792	0.0766
	(0.1429)	(0.0581)	(0.0650)	(0.0566)	(0.1766)	(3.631)	(0.0741)
Midline Treatment Difference	0.0465	0.0140	-0.0225	-0.1156**	-0.0241	-3.873	0.0332
	(0.1438)	(0.0571)	(0.0666)	(0.0569)	(0.1624)	(3.611)	(0.0754)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes	No	No
Observations	3,732	3,690	3,721	3,705	3,732	3,732	3,732
Mean of DV	0.023	-0.007	-0.003	0.019	0.015	21.275	0.726

Notes: The summary stats provided correspond to Panel C: Difference (High Exposure - Low Exposure)
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Impacts of Rainfall Shocks by Regional Exposure

Panel A: Aggregate Rainfall Impacts					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Exposed to Rainfall Shock	0.1438	0.0048	0.0078	0.0136	0.0052
	(0.1208)	(0.0430)	(0.0447)	(0.0436)	(0.1127)
Panel B: Rainfall Impacts on High Exposure Region Subsample					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Exposed to Rainfall Shock	0.0413	0.0287	0.0461	-0.0065	0.0539
	(0.0893)	(0.0327)	(0.0331)	(0.0364)	(0.0911)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes
Observations	3,865	3,810	3,843	3,801	3,865
Mean of DV	0.063	-0.018	-0.002	0.022	0.074

Notes: The summary stats provided correspond to Panel B: Rainfall Impacts on High Exposure Region Subsample
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Summary statistics, by survey timing relative to the storm

Variable	Surveyed Post-Storm	Surveyed Pre-Storm	Difference (Post-Storm − Pre-Storm)
Outcomes			
CC quiz (BL, standardized)	-0.067	0.029	-0.096 (0.097)
CC skepticism (BL, standardized)	0.019	-0.003	0.022 (0.038)
CC engagement (BL, standardized)	-0.029	0.009	-0.038 (0.040)
CC anxiety (BL, standardized)	0.049	-0.026	0.075** (0.037)
Math score (BL, standardized)	0.214	-0.131	0.345*** (0.108)
Donation amount (EL)	22.289	20.595	1.694 (2.002)
Baseline covariates			
Female	0.643	0.597	0.045 (0.052)
High climate exposure	0.959	0.244	0.715*** (0.060)
Student age	13.622	13.813	-0.191*** (0.061)
Mother secondary education	0.592	0.679	-0.087** (0.035)
Father secondary education	0.595	0.647	-0.052 (0.032)
Asset index, (mean zero)	-0.245	0.149	-0.394*** (0.097)
Teacher climate knowledge	0.819	0.835	-0.016 (0.012)
Teacher climate anxiety	0.555	0.542	0.013 (0.017)
Teacher is female	0.179	0.319	-0.140*** (0.044)
Teacher age	43.081	43.053	0.027 (0.779)
Teacher has M.A. degree	0.556	0.597	-0.041 (0.057)
Observations	1985	3503	

Notes: The first two columns report means for students surveyed after and before the storm, respectively. The difference column reports Mean(Post-Storm) − Mean(Pre-Storm), estimated from separate regressions of each variable on an indicator for being surveyed after the storm. Standard errors are clustered at the school level. Significance levels: $*p < 0.10$, $**\,p < 0.05$, $***p < 0.01$.

Table A7: Joint Education and Rainfall Treatment Effects

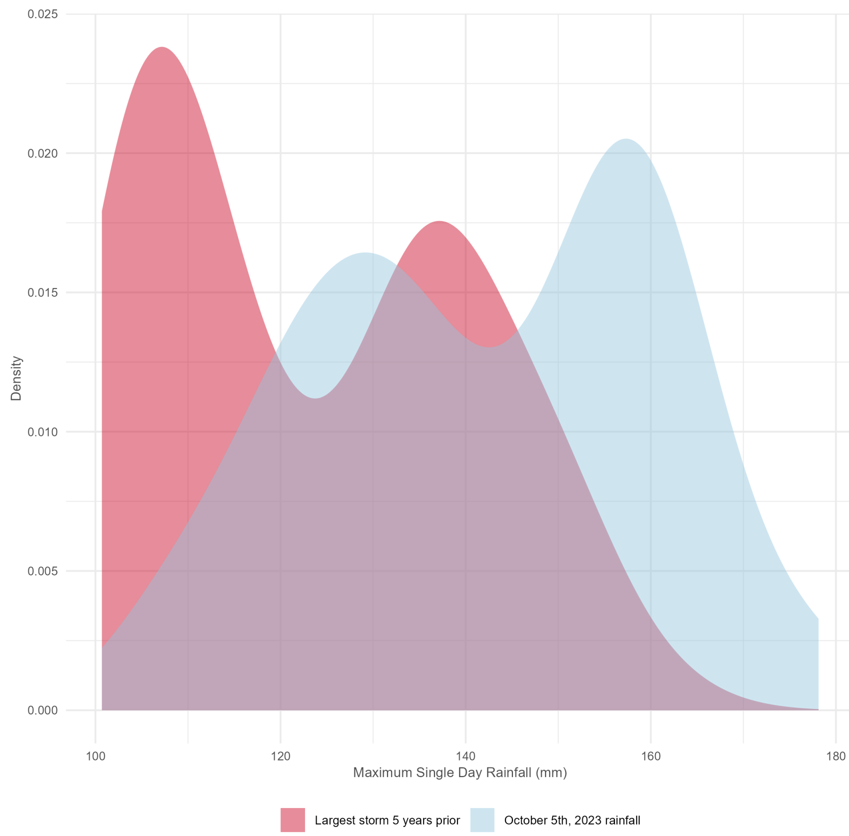
Panel A: Education Impacts BL–ML					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Baseline Treatment	0.0132	0.0242	-0.0317	-0.0018	-0.0703
	(0.0712)	(0.0307)	(0.0293)	(0.0304)	(0.0769)

Panel B: Rainfall Impacts BL–ML					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Exposed to Rainfall Shock	0.1438	0.0048	0.0078	0.0136	0.0052
	(0.1208)	(0.0430)	(0.0447)	(0.0436)	(0.1127)

Panel C: Interaction of Education and Rainfall Treatments					
Dependent Variables:	Climate Knowledge Score	Climate Skepticism Index	Climate Anxiety	Problem Solving Index	Math Quiz Score
Variables					
Baseline Treatment	0.0520	0.0706*	-0.0372	-0.0189	-0.0950
	(0.0694)	(0.0416)	(0.0351)	(0.0380)	(0.0904)
Exposed to Rainfall Shock	0.2189	0.0454	0.0306	-0.0092	-0.0050
	(0.1581)	(0.0631)	(0.0606)	(0.0573)	(0.1710)
Rain Shock and Treatment	-0.2126	-0.1214	0.0024	0.0399	0.0554
	(0.1757)	(0.0734)	(0.0666)	(0.0748)	(0.1946)
Baseline X Controls	Yes	Yes	Yes	Yes	Yes
Baseline Y Controls	Yes	Yes	Yes	Yes	Yes
Observations	3,865	3,810	3,843	3,801	3,865
Mean of DV	0.063	-0.018	-0.002	0.022	0.074

Notes: The summary stats provided correspond to Panel C: Interaction of Education and Rainfall Treatments
Standard-errors clustered at the school-level are shown in parentheses. Signif. Codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A2: Distribution of daily rainfall and relative frequency of extreme events



Note: This figure plots the distribution of the single-day rainfall observed at each school on October 5, 2023, and the maximum single-day rainfall observed at the same schools during the five years prior to the study. Each observation corresponds to one school (N = 140). The densities are smoothed kernel estimates to facilitate comparison of the distribution of extreme rainfall events across periods.